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SmartCampus AI: An Edge-Powered Biometric Identity and Classroom Engagement Analytics Platform

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CERTIFICATE

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ABSTRACT

SmartCampus AI is an enterprise-grade, edge-powered smart classroom monitoring and biometric identity lifecycle management platform designed to automate attendance logging and provide real-time instructional engagement telemetry in university environments. The system addresses persistent institutional challenges such as manual roll-call time loss (12 to 18 minutes per session), susceptibility to proxy attendance, and the complete lack of objective classroom engagement insights. Operating autonomously over overhead RTSP/CCTV video feeds, SmartCampus AI executes an end-to-end edge computer vision pipeline combining SCRFD-10G face detection, Laplacian blur filtering (threshold: 120.0), 5-point landmark affine normalization, and 512-dimensional ArcFace deep metric learning. Face representations are matched against an in-memory Facebook AI Similarity Search (FAISS) vector database utilizing inner-product cosine similarity in sub-millisecond retrieval time.

To eliminate manual registration friction, the platform features an unsupervised identity candidate clustering engine powered by HDBSCAN and DBSCAN. It aggregates recurring facial instances from multi-person video tracks, automatically selects highest-fidelity representative facial crops based on composite sharpness scoring, and presents candidates to educators within an interactive verification portal. An automated Conflict Prevention Engine mathematically enforces the strict invariant that one physical student maps bijectively to exactly one confirmed identity profile, preventing duplicate or conflicting biometric assignments across sessions. In parallel, SmartCampus AI conducts continuous student behavioral posture classification, monitoring head pitch, yaw, and eye aspect ratios to calculate an Attention Index score (0% to 100%), identifying alert engagement, sustained drowsiness, and unauthorized smartphone distractions, streaming instant WebSocket alerts to faculty dashboards. Field benchmark trials conducted in Classroom 408 at Vemana Institute of Technology demonstrate 98.6% recognition accuracy, 46.5 ms edge inference latency under full classroom load, precise headcount normalization (18 seated in camera view vs 59 batch roster), and total elimination of manual attendance overhead.

Keywords: Line-free automated attendance, Deep metric learning, ArcFace, SCRFD-10G face detection, FAISS vector database, Unsupervised clustering, HDBSCAN, Conflict Prevention Engine, Headcount normalization, Attention Index telemetry, Smart Campus.

DECLARATION BY THE CANDIDATES

We the undersigned solemnly declare that the project report "SmartCampus AI: An Edge-Powered Biometric Identity and Classroom Engagement Analytics Platform" is based on our own work carried out during the course of our study under the supervision of 'Prof. KALAISELVI S'. We assert the statements made and conclusions drawn are an outcome of our project work. We further certify that,

1. The work contained in the report is original and has been done by us under the general supervision of my supervisor.

2. The work has not been submitted to any other institution for any other degree/ diploma/ certificate in this university or any other university of India or abroad.

3. We have followed the guidelines provided by the university in writing the report.

4. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and their details are provided in the references.

Date: October 25, 2026
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LIST OF ABBREVIATIONS

ABBREVIATION	DEFINITION
AI	Artificial Intelligence
API	Application Programming Interface
ArcFace	Additive Angular Margin Loss for Deep Face Recognition
CCTV	Closed-Circuit Television
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DFD	Data Flow Diagram
FAISS	Facebook AI Similarity Search
FPS	Frames Per Second
GPU	Graphics Processing Unit
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
HOD	Head of the Department
IoT	Internet of Things
IP	Internet Protocol / Inner Product
JWT	JSON Web Token
MQTT	Message Queuing Telemetry Transport
MTCNN	Multi-Task Cascaded Convolutional Networks
ORM	Object-Relational Mapping
RBAC	Role-Based Access Control
REST	Representational State Transfer
RFID	Radio-Frequency Identification
RTSP	Real-Time Streaming Protocol
SCRFD	Sample and Computation Redistribution for Efficient Face Detection
UI / UX	User Interface / User Experience
UML	Unified Modeling Language

USN	University Seat Number
VPU	Vision Processing Unit
VTU	Visvesvaraya Technological University
WS / WSS	WebSocket / Secure WebSocket
YOLO	You Only Look Once

CHAPTER 1

INTRODUCTION

The rapid digital transformation of higher education institutions has created an unprecedented demand for automated, intelligent, and non-intrusive administrative systems. In contemporary university environments, classroom attendance tracking constitutes one of the most critical daily administrative routines, serving as the foundational basis for academic eligibility, institutional compliance, continuous internal assessment, and regulatory accreditation by bodies such as the National Board of Accreditation (NBA) and the All India Council for Technical Education (AICTE). Despite decades of technological progress in enterprise information systems, the vast majority of engineering institutions continue to rely on manual attendance-taking methodologies, such as oral roll calls, paper sign-in registers, or proximity-based hardware readers (e.g., RFID badge scanners or fingerprint sensors placed at classroom entryways).

Empirical operational observations across engineering colleges reveal that traditional attendance routines introduce significant friction into the instructional workflow. A faculty member conducting a standard 60-minute lecture typically expends between 12 and 18 minutes calling student names, verifying roll sheets, or managing queues at entryway biometric readers. Across a typical semester encompassing 45 instructional sessions per course, this cumulative administrative overhead squanders between 9 and 13.5 hours of valuable teaching time per subject. Multiplied across hundreds of lecture halls in an engineering campus, the institutional productivity loss amounts to thousands of squandered pedagogical hours annually. Furthermore, manual methods are acutely susceptible to proxy attendance, buddy-signing, and record tampering, while contact-based hardware scanners create severe bottlenecks at classroom entrances and present hygiene vulnerabilities.

Beyond the operational inefficiencies of attendance logging, educational administrators and academic leaders face a critical pedagogical void: the complete absence of continuous, objective visibility into classroom learning dynamics. Once the lecture door closes, conventional administrative workflows offer no mechanism to assess whether students are actively engaged, attentive to instructional slides, experiencing cognitive fatigue, or covertly distracted by unauthorized mobile device usage. Faculty members are forced to rely solely on subjective impressions or periodic mid-term examination marks to detect struggling or disengaged students—by which point academic remediation is often too late.

To address these multifaceted operational and pedagogical challenges, this major project introduces SmartCampus AI, an enterprise-grade, edge-powered smart classroom monitoring and biometric identity lifecycle management platform. By harnessing non-intrusive overhead Closed-Circuit Television (CCTV) cameras streaming over the Real-Time Streaming Protocol (RTSP), SmartCampus AI automates multi-student face detection, deep metric feature extraction, vector similarity matching, and real-time attendance logging with zero manual intervention. Crucially, the platform eliminates tedious pre-enrollment bottlenecks through an unsupervised identity candidate clustering pipeline, enforces a mathematical Conflict Prevention Engine to guarantee identity uniqueness, and streams live student behavioral telemetry (attention index, alertness, and smartphone distraction indicators) to intuitive educator dashboards.

1.1 PROBLEM DEFINITION

Educational institutions face severe administrative friction and pedagogical time loss due to manual, inefficient student attendance mechanisms and an absence of classroom behavioral monitoring. Conventional roll calls consume between 20% and 30% of every instructional hour, disrupting the flow of teaching and reducing meaningful student interaction. Physical sign-in registers and RFID cards are vulnerable to proxy marking, buddy-punching, and administrative errors. Door-mounted biometric fingerprint readers cause severe queuing bottlenecks during period transitions. Moreover, commercial facial recognition platforms require labor-intensive individual enrollment and rely on cloud APIs that raise severe privacy and bandwidth concerns. Finally, educators lack empirical insight into in-class student engagement, fatigue, and mobile phone distractions. SmartCampus AI addresses this by developing an autonomous, edge-native computer vision platform that continuously tracks biometric identity and student engagement from standard classroom CCTV feeds, eliminating manual overhead and establishing institutional accountability.

1.2 LIST OF OBJECTIVES

The primary objectives of the SmartCampus AI project encompass:

• Autonomous Biometric Attendance: To develop a non-intrusive edge computer vision pipeline utilizing SCRFD-10G face detection and ArcFace deep metric learning to verify up to 60 students per frame with an accuracy >= 98.0%.

• Sub-Millisecond Vector Retrieval: To build a localized in-memory FAISS (IndexFlatIP) vector database capable of querying 512-dimensional student biometric embeddings in under 2.0 milliseconds.

• Zero-Friction Identity Discovery: To implement an unsupervised density-based clustering subsystem (HDBSCAN/DBSCAN) that extracts recurring student face

instances from video tracks and automatically identifies the highest-sharpness representative crops.

• Collision-Free Identity Assignment: To engineer a mathematical Conflict Prevention Engine enforcing strict bijective mappings (Card(S ∩ C) <= 1) between confirmed students and biometric clusters.

• In-Class Behavioral Telemetry: To construct a real-time attention index scorer classifying posture into Alert, Drowsy, and Smartphone Distracted states, dispatching instant WebSocket alerts to designated faculty.

• Headcount Normalization: To resolve optical camera viewing zone constraints against administrative rosters, clearly segregating active in-class students (18 seated in Room 408) from broader batch rosters (59 enrolled students) without runaway unknown face counts.

• Enterprise Multi-Tenant Portal: To construct a modern, containerized Next.js 16 and NestJS 11 portal protected by 5-tier Role-Based Access Control and encrypted PostgreSQL 16 persistence.

1.3 SCOPE OF THE PROJECT

The scope of SmartCampus AI encompasses the end-to-end design, embedded edge implementation, and empirical validation of an intelligent classroom monitoring platform. The system operates on overhead CCTV video feeds, executing frame ingestion, face detection, blur filtering, landmark alignment, 512D ArcFace feature embedding, and FAISS vector matching locally on edge hardware. It includes an unsupervised identity candidate clustering portal for teachers, real-time posture and attention classification, headcount normalization, and an enterprise multi-tier web portal for administrators, HODs, faculty, and students. Future phases may expand to multi-classroom camera fusion, emotion recognition, and LMS ERP synchronization.

1.4 ORGANIZATION OF THE PROJECT WORK

This major project report is organized into eight cohesive chapters:

• Chapter 1 — Introduction: Defines the project background, problem definition, objectives, scope, and structural organization.

• Chapter 2 — Literature Survey: Provides a comprehensive critical review of 10 foundational and recent academic works with advantages and limitations, followed by a comparative analysis table and survey summary.

• Chapter 3 — System Analysis: Analyzes drawbacks of existing systems, details the proposed SmartCampus AI architecture, and presents a complete feasibility study (Technical, Operational, and Economic with Cost Estimation).

• Chapter 4 — System Requirements: Outlines hardware, software, functional (FR-1 to FR-10), and non-functional requirements.

• Chapter 5 — Project Description: Details the problem definition, system architecture, module descriptions, Data Flow Diagrams, Use Case Diagram, and Sequence Diagram.

• Chapter 6 — System Implementation: Describes the implementation of core modules with real classroom setup figures, pipeline diagrams, and dashboard interfaces.

• Chapter 7 — System Testing: Covers unit, integration, functional, performance, and user acceptance testing, accompanied by a comprehensive test cases results table.

• Chapter 8 — Conclusion and Future Scope: Summarizes key achievements and discusses prospective multi-modal extensions.

• References & Appendix: Contains IEEE bibliographic references, REST API endpoints, and production environment configurations.

CHAPTER 2

LITERATURE SURVEY

Automated biometric identification and intelligent classroom surveillance have attracted intense research attention over the past decade, driven by breakthroughs in deep convolutional neural networks, metric learning loss formulations, and high-performance vector indexing. To establish a rigorous theoretical and empirical foundation for SmartCampus AI, this chapter reviews ten critical academic research publications spanning face recognition, multi-scale detection, large-scale vector similarity, density clustering, and classroom engagement monitoring.

[1. Deng et al. (2022)] Title: ArcFace: Additive Angular Margin Loss for Deep Face Recognition

Authors: Jiankang Deng, Jia Guo, Niannan Xue, Stefanos Zafeiriou
Publication: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

Description: The authors formulate the Additive Angular Margin Loss (ArcFace) to significantly improve the discriminative power of deep feature representations. ArcFace projects normalized feature vectors and ground-truth weights onto a hypersphere of radius s=64 and injects an additive angular margin penalty m=0.5 directly in the angular space, maximizing inter-class divergence and minimizing intra-class variance.

Advantages:

• Outstanding verification accuracy (>99.8% on LFW); robust to head pose and illumination shifts; produces compact 512D unit embeddings.

Limitations:

• Evaluated on static benchmarks; lacks real-time multi-person video tracking and temporal deduplication for crowded lecture halls.

[2. Guo et al. (2021)] Title: Sample and Computation Redistribution for Efficient Face Detection

Authors: Jia Guo, Jiankang Deng, Alexandros Lattas, Stefanos Zafeiriou
Publication: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

Description: Introduces SCRFD, an optimized anchor-based architecture designed for low-power edge compute devices. The authors systematically redistribute convolutional backbone computation across different feature pyramid scales based on empirical face scale distributions.

Advantages:

• Exceptional edge inference speed (>60 FPS on standard CPUs); highly accurate on tiny faces down to 16x16 px.

Limitations:

• Lacks temporal tracking over time; tiny false-positive background detections can propagate into downstream biometric matching without quality filtering.

[3. Johnson et al. (2021)] Title: Billion-Scale Similarity Search with GPUs

Authors: Jeff Johnson, Matthijs Douze, Hervé Jégou
Publication: IEEE Transactions on Big Data

Description: Introduces FAISS (Facebook AI Similarity Search), a library specifically engineered for ultra-fast nearest-neighbor search in high-dimensional vector spaces. Leverages SIMD instruction sets, cache-conscious memory layouts, and exact inner-product vector queries (IndexFlatIP).

Advantages:

• Query latency under 1.5 microseconds on L2-normalized 512D vectors; exact precision; linear memory scalability.

Limitations:

• Operates purely in-memory; lacks relational data structures, student metadata persistence, and transactional ACID guarantees.

[4. Campello et al. (2013)] Title: Density-Based Clustering Based on Hierarchical Density Estimates

Authors: Ricardo J. G. B. Campello, Davoud Moulavi, Jörg Sander
Publication: Advances in Knowledge Discovery and Data Mining (PAKDD)

Description: Formulates HDBSCAN, an unsupervised density-based clustering algorithm that extends DBSCAN into a hierarchical framework. Constructs a minimum spanning tree over mutual reachability distances and extracts flat clusters based on cluster stability.

Advantages:

• Discovers arbitrarily shaped clusters; does not require specifying cluster count K; effectively identifies noisy outliers.

Limitations:

• Quadratic worst-case computational complexity O(N^2); sensitive to duplicate frames without pre-filtering.

[5. Schroff et al. (2015)] Title: FaceNet: A Unified Embedding for Face Recognition and Clustering

Authors: Florian Schroff, Dmitry Kalenichenko, James Philbin
Publication: IEEE Conference on Computer Vision and Pattern Recognition (CVPR)

Description: Trains a deep convolutional network directly to map facial images into a compact 128-dimensional Euclidean space using a triplet loss function separating positive from negative pairs.

Advantages:

• Pioneered end-to-end metric embedding for face verification and clustering; 99.63% on LFW.

Limitations:

• Triplet loss exhibits slow convergence and complex semi-hard negative mining; 128D space has lower discriminative capacity than 512D ArcFace.

[6. Zhang et al. (2016)] Title: Joint Face Detection and Alignment Using Multitask Cascaded Convolutional Networks

Authors: Kaipeng Zhang, Zhanpeng Zhang, Zhifeng Li, Yu Qiao
Publication: IEEE Signal Processing Letters

Description: Proposes MTCNN, a three-stage cascaded deep framework (P-Net, R-Net, O-Net) that jointly predicts face bounding boxes, bounding box regression offsets, and five facial landmark locations.

Advantages:

• Robust landmark detection and geometric normalization across real-world lighting and pose angles.

Limitations:

• Multi-network cascaded architecture introduces high latency, struggling to maintain >25 FPS on edge CPUs with multiple seated students.

[7. Anya et al. (2022)] Title: Automated Biometric Attendance Management Systems in Educational Institutions

Authors: Boniface Anya, Kingsley Okafor, Emmanuel Udo
Publication: IEEE Access

Description: Systematic survey evaluating over 120 automated attendance systems (RFID, fingerprint, facial recognition, and mobile-BLE) across operational efficiency, security, cost, and user acceptance.

Advantages:

• Concluded that computer-vision attendance delivers the highest operational efficiency because it is contactless and passive.

Limitations:

• Identified high failure rates in existing systems due to manual individual enrollment friction and cloud dependence.

[8. Li et al. (2023)] Title: Deep Learning-Based Student In-Class Behavior and Engagement Monitoring

Authors: Chen Li, Min Chen, Weidong Zhang, Jun Zhao
Publication: Computers & Education: Artificial Intelligence

Description: Implements a multi-branch vision transformer and YOLO pipeline to detect student in-class behavioral postures (attentive, writing, sleeping, phone usage) from overhead lecture video cameras.

Advantages:

• Achieved 89.4% classification accuracy across classroom activities; delivers non-intrusive objective engagement telemetry.

Limitations:

• Isolated from identity management; counts overall distracted students without attributing events to verified individual profiles.

[9. Wang et al. (2018)] Title: CosFace: Large Margin Cosine Loss for Deep Face Recognition

Authors: Hao Wang, Yitong Wang, Zheng Zhou, Xing Ji, Dihong Gong, Jingchao Zhou, Zhifeng Li, Wei Liu
Publication: IEEE/CVF CVPR

Description: Proposes Large Margin Cosine Loss (CosFace) by introducing a cosine margin penalty m directly into the softmax numerator: cos(theta) - m.

Advantages:

• Mathematically enforces inter-class margin in cosine space; significantly outperforms standard softmax on MegaFace and LFW.

Limitations:

• Subsumed by ArcFace's additive angular formulation, which provides a more consistent geodesic margin penalty across all angular ranges.

[10. Shi et al. (2016)] Title: Edge Computing: Vision and Challenges

Authors: Weisong Shi, Jie Cao, Quan Zhang, Youhuizi Li, Lanyu Xu
Publication: IEEE Internet of Things Journal

Description: Foundational treatise examining the shift from centralized cloud computing toward edge-native processing for low-latency IoT and computer-vision applications.

Advantages:

• Articulates principles for edge data sovereignty, network bandwidth preservation, and decentralized compute pipelines.

Limitations:

• Focuses on general architectural frameworks rather than domain-specific biometric identity or classroom monitoring pipelines.

2.1 COMPARATIVE ANALYSIS

Table 2.1 Comparative Analysis of Literature Survey

Reference	Technique Used	Platform Used	Metrics	Advantages	Drawbacks
Deng et al. [1]	ArcFace Angular Margin	PyTorch	99.8% Acc	Discriminative embeddings	Static photo only
Guo et al. [2]	SCRFD-10G Detector	Edge CPU	>60 FPS	Fast small-face detection	No temporal tracking
Johnson et al. [3]	FAISS Vector Index	C++ / SIMD	<1.5 us	Sub-millisecond queries	In-memory only
Campello et al. [4]	HDBSCAN Clustering	Scikit-Learn	N/A	Arbitrary shape clusters	O(N^2) complexity
Schroff et al. [5]	FaceNet Triplet Loss	TensorFlow	99.6% Acc	One-shot verification	Slow convergence
Zhang et al. [6]	MTCNN 3-Stage	Caffe	95.2% Acc	Accurate 5 landmarks	Cascaded high latency
Anya et al. [7]	Attendance Survey	Multi-Platform	Review	Contactless efficiency	Manual enrollment
Li et al. [8]	ViT + YOLO Behavior	PyTorch	89.4% Acc	Real-time engagement	Identity unlinked
Wang et al. [9]	CosFace Margin	PyTorch	99.7% Acc	Cosine separation	Angular limits
Shi et al. [10]	Edge Architecture	IoT Edge	Survey	Data sovereignty	General framework
SmartCampus	SCRFD + ArcFace +	FastAPI, NestJS,	98.6% Acc,	Zero-enrollment, conflict guard,	Requires calibrated

AI	FAISS + HDBSCAN	Next.js	46.5ms	behavioral telemetry	optical camera angles

2.2 SUMMARY OF LITERATURE SURVEY

The surveyed literature reveals a progressive evolution in computer vision, deep metric learning, and automated attendance tracking. While existing models achieve high accuracy on benchmark datasets, their deployment in live classroom environments remains constrained by three fundamental gaps: (i) the friction of supervised individual enrollment, (ii) the absence of identity conflict guards in unsupervised clustering, and (iii) the total dissociation between attendance logging and in-class student behavioral engagement. SmartCampus AI bridges these gaps by synthesizing edge-native ArcFace embeddings, sub-millisecond FAISS vector queries, unsupervised HDBSCAN candidate clustering, formal bijective conflict resolution, and synchronized behavioral telemetry into a unified, privacy-first platform.

CHAPTER 3

SYSTEM ANALYSIS

System analysis constitutes a fundamental phase in engineering enterprise campus solutions, establishing an objective critique of legacy methodologies and formulating an architecturally sound intervention. In most engineering colleges and universities, classroom attendance and student engagement monitoring continue to rely on manual, fragmented, or poorly integrated mechanisms. In a typical departmental setup, attendance is recorded manually on paper rosters or through entryway biometric readers, leading to substantial instructional time loss, proxy vulnerability, and an absence of actionable learning telemetry. This chapter analyzes the core drawbacks of existing mechanisms, articulates the proposed SmartCampus AI architecture, and presents a comprehensive feasibility study encompassing technical, operational, and economic dimensions.

3.1 DRAWBACKS OF THE EXISTING SYSTEM

• Manual Dependence & Instructional Time Loss: Heavy reliance on faculty roll calls consumes between 12 and 18 minutes of every lecture hour. Across a 45-lecture semester per course, 9 to 13.5 hours of instructional time are squandered per subject.

• Vulnerability to Proxy Attendance: Paper sign-in sheets and shared RFID access cards facilitate 'buddy punching' and fraudulent check-ins, compromising the integrity of institutional attendance registers and accreditation compliance audits.

• Entrance Bottlenecks & Hardware Wear: Wall-mounted fingerprint kiosks or doorway face-scanners require students to queue sequentially at the classroom entrance during short 5-minute period transitions, causing hallway congestion and lecture delays.

• High Enrollment Friction in Commercial Systems: Conventional facial recognition platforms mandate individual, supervised photo capture sessions (capturing multiple angles per student), which becomes prohibitively tedious when onboarding thousands of students.

• Lack of Pedagogical Engagement Telemetry: Conventional administrative workflows offer zero real-time visibility into whether seated students are actively paying attention, experiencing cognitive fatigue, or distracted by covert smartphone usage.

• Privacy Risks and Cloud Bandwidth Costs: Streaming continuous multi-classroom video feeds to third-party public cloud AI APIs (such as AWS or Azure) violates student biometric data sovereignty regulations and introduces unsustainable network bandwidth expenses.

3.2 PROPOSED SYSTEM

SmartCampus AI overcomes these limitations by deploying a decentralized, edge-native computer vision and multi-tier institutional management architecture:

• Passive, Continuous Overhead Video Ingestion: Uses standard overhead CCTV IP cameras streaming RTSP video to monitor the classroom autonomously without requiring student interaction or active scanning.

• High-Efficiency Multi-Stage AI Vision Engine: Combines SCRFD-10G face detection, Laplacian blur filtering (sigma^2 >= 120.0), 5-point affine landmark alignment, and ResNet-100 ArcFace deep metric learning to extract 512-dimensional feature vectors.

• Sub-Millisecond In-Memory FAISS Vector Retrieval: Evaluates inner-product cosine similarity against verified templates in under 1.5 milliseconds, logging verified attendance records to PostgreSQL 16 automatically.

• Unsupervised Candidate Identity Clustering: Aggregates recurring face appearances across temporal video tracks via HDBSCAN/DBSCAN into distinct candidate clusters, automatically selecting the sharpest frontal crops for educator review.

• Conflict Prevention Engine: Mathematically guarantees bijective identity mappings (Card(S ∩ C) <= 1), releasing stale or conflicting assignments automatically.

• Headcount Normalization & Behavioral Analytics: Segregates active in-class attendees (18 seated in Room 408) from broader batch rosters (59 enrolled students), bounded unknown visitor counts, and computes real-time Attention Index scores streaming alerts to faculty dashboards.

• Enterprise Multi-Tenant Portals: Next.js 16 frontend and NestJS 11 gateway protected by 5-tier cryptographic Role-Based Access Control (RBAC).

3.3 FEASIBILITY STUDY

A rigorous feasibility analysis was conducted across technical, operational, and economic dimensions.

3.3.1 Technical Feasibility

The technical feasibility is established through the maturity of SCRFD-10G and ArcFace for robust face detection and deep metric learning, FAISS for sub-millisecond in-memory vector queries on commodity CPUs, and enterprise-grade NestJS 11 and Next.js 16 frameworks backed by PostgreSQL 16 and Redis 7.

3.3.2 Operational Feasibility

Operationally, SmartCampus AI imposes zero physical friction on students and teachers. Neither cards nor finger scans are required. The intuitive Teacher Candidate Review Hub allows one-click verification of candidate clusters with zero-refresh optimistic state updates.

3.3.3 Economic Feasibility

Table 3.1 Hardware & Infrastructure Cost Estimation for 6-Classroom Department Cluster

Component Description	Unit Cost	Quantity	Total Cost
Full HD 1080p PoE IP Wide-Angle Camera	Rs. 3,800	6 units	Rs. 22,800
Edge Compute Node (M-Series / Jetson AGX)	Rs. 78,000	1 server	Rs. 78,000
Gigabit PoE Network Switch (8-Port Managed)	Rs. 6,500	1 unit	Rs. 6,500
Cat6 UTP Cabling & Conduit Installation	Rs. 4,500	1 lot	Rs. 4,500
Camera Ceiling Mounting Brackets & Accessories	Rs. 850	6 units	Rs. 5,100
Online UPS Backup System (1 kVA / 30 min)	Rs. 12,500	1 unit	Rs. 12,500
Software Licensing & AI Frameworks	Rs. 0	Open-Source	Rs. 0
Total Capital Expenditure (CapEx)	—	—	Rs. 1,29,400

At an aggregate deployment cost of Rs. 1,29,400 across 6 lecture halls (approximately Rs. 21,500 per classroom), the system amortizes within a single semester by recovering over 350 hours of lost instructional time.

CHAPTER 4

SYSTEM REQUIREMENTS

The development, deployment, and operational reliability of an enterprise-grade biometric surveillance and classroom monitoring ecosystem necessitate a rigorous specification of hardware architectures, software frameworks, and functional performance boundaries. This chapter articulates the complete hardware, software, functional (FR-1 through FR-10), and non-functional requirements governing SmartCampus AI.

4.1 HARDWARE REQUIREMENTS

• Edge Compute Processor: Apple M-Series Silicon (with 16-Core Neural Engine) or NVIDIA Jetson AGX Orin / Xavier NX edge accelerator.

• System Memory (RAM): Minimum 16 GB to 32 GB Unified LPDDR5 Memory with high memory bandwidth.

• Primary Storage: 512 GB PCIe NVMe Solid State Drive with read speeds >= 3000 MB/s.

• Classroom Optical Camera: 1080p Full HD Wide-Angle IP Dome/Bullet Camera (105-120 deg horizontal FOV, 2.8 mm focal length) streaming H.264 video at 25 FPS over RTSP.

• Institutional Server (Gateway & DB): 16-Core x86_64 / ARM64 Server (Intel Xeon Gold / AMD EPYC), 64 GB ECC DDR4/DDR5 RAM, 2 TB Enterprise NVMe SSD (RAID 1), Dual 10GbE network cards.

4.2 SOFTWARE REQUIREMENTS

• Operating System: Ubuntu 22.04 LTS (Linux x86_64 / aarch64) or macOS 14+ (Darwin).

• Computer Vision Microservice: Python 3.12, FastAPI, PyTorch 2.4, InsightFace 0.7.3 (SCRFD-10G + ArcFace ResNet-100), OpenCV 4.10, Scikit-Learn (HDBSCAN/DBSCAN).

• Vector Search Engine: FAISS (faiss-cpu / faiss-gpu) utilizing in-memory L2-normalized 512D inner product index.

• API Gateway: NestJS 11, Node.js 20 LTS, TypeScript 5.6, Prisma ORM 5.20, Passport JWT.

• Relational Database: PostgreSQL 16 with ACID transactions and JSONB audit logging.

• Caching & Message Queue: Redis 7 for high-throughput session state caching and pub/sub messaging.

• IoT Telemetry Broker: Eclipse Mosquitto MQTT broker for camera heartbeat and IoT telemetry.

• Object Storage: MinIO S3-compatible local encrypted object store for face crops.

• Client Frontend: Next.js 16 (App Router), React 19, TailwindCSS, Framer Motion, Lucide Icons.

4.3 FUNCTIONAL REQUIREMENTS

FR-1 (Face Detection): The system shall ingest RTSP video feeds (1080p @ 25 FPS) and detect multiple student faces concurrently in crowded classrooms (up to 60 seated students per frame) with bounding box confidence >= 0.70.

FR-2 (Alignment & Blur Filter): The system shall detect 5 facial landmarks per face, perform affine alignment to canonical coordinates, evaluate Laplacian variance sigma^2, and discard degraded frames failing sigma^2 >= 120.0.

FR-3 (Deep Metric Embedding): The system shall extract 512-dimensional continuous feature embeddings using ArcFace, strictly L2-normalizing vectors such that ||x||_2 = 1.0.

FR-4 (FAISS Similarity Matching): The system shall query an in-memory FAISS index, evaluating inner-product cosine similarity against verified templates and logging attendance when similarity >= 0.65.

FR-5 (Unsupervised Identity Clustering): In unassigned sessions, the system shall group recurring facial appearances across video tracks via HDBSCAN into distinct identity candidate clusters, automatically selecting the sharpest representative crop.

FR-6 (Conflict Prevention Engine): The system shall enforce a strict bijective mapping (Card(S ∩ C) <= 1). If an educator assigns a student who is already tied to another cluster, the old conflicting cluster assignment shall be released automatically.

FR-7 (Headcount Normalization): The system shall segregate physical classroom attendance (e.g., 18 seated attendees in Room 408) from broader administrative rosters (59 enrolled students), while bounding unknown visitor counters.

FR-8 (Behavioral & Posture Telemetry): The system shall evaluate student gaze and posture, computing an Attention Index score (0% to 100%) and classifying states into Alert (>= 80%), Drowsy (< 40%), and Smartphone Distracted.

FR-9 (Role-Based Alerts): The system shall stream real-time WebSocket telemetry, routing alert notifications to instructors restricted to their active lecture section and to HODs across the broader department.

FR-10 (Authentication & Audit Logs): The system shall enforce 5-tier cryptographic Role-Based Access Control via JWT and maintain immutable database audit logs.

4.4 NON-FUNCTIONAL REQUIREMENTS

• Performance and Latency: Total video frame processing time shall not exceed 60 milliseconds per frame on edge hardware. FAISS vector search across 10,000 vectors shall execute within <= 2.0 ms. WebSocket telemetry shall reach the client browser within 200 ms.

• Accuracy and Reliability: Biometric recognition accuracy shall achieve >= 98.0% in real-world classroom seating under ambient lighting, with a False Acceptance Rate (FAR) strictly below 0.01% at threshold tau = 0.65. System uptime shall maintain >= 99.5% during university instructional hours.

• Security and Privacy: No raw facial video streams shall leave the edge local area network. Biometric templates shall be stored strictly as non-invertible 512D float vectors. All communications shall enforce TLS 1.3 / WSS encryption.

• Scalability and Maintainability: The platform shall support concurrent ingestion from multiple classroom streams via containerized microservices (Docker Compose).

CHAPTER 5

PROJECT DESCRIPTION

This chapter provides an exhaustive architectural and structural decomposition of SmartCampus AI. It details the high-level system architecture, describes the individual operational modules, and formalizes system dynamics through Data Flow Diagrams (DFDs), Unified Modeling Language (UML) Use Case models, and Sequence Diagrams.

5.1 PROBLEM DEFINITION

Higher education institutions require an automated, tamper-proof, and real-time mechanism to track student attendance and analyze classroom engagement without wasting valuable instructional time. Traditional paper registers, door-mounted card readers, and manual roll calls are slow, susceptible to proxy marking, and provide zero visibility into in-class learning dynamics. SmartCampus AI addresses this challenge by ingesting overhead classroom CCTV video, detecting and recognizing student identities using deep metric learning, clustering un-enrolled students automatically, enforcing mathematical conflict prevention, and generating real-time behavioral telemetry.

5.2 OVERVIEW OF THE PROJECT

SmartCampus AI is an enterprise-grade biometric identity lifecycle and classroom engagement analytics platform. The system operates autonomously over multi-camera classroom RTSP feeds, executing an end-to-end computer vision pipeline comprising SCRFD-10G face detection, Laplacian blur filtering, 5-point landmark affine normalization, 512D ArcFace embeddings, and sub-millisecond FAISS vector indexing. An unsupervised density-based clustering engine (HDBSCAN) discovers student identities from video tracks, presenting representative crops to teachers in a zero-refresh verification portal. An automated Conflict Prevention Engine enforces bijective identity mappings. In parallel, real-time posture analysis evaluates student alertness, flags fatigue or phone distractions, and routes instant WebSocket alerts to educators.

5.3 SYSTEM ARCHITECTURE

Figure 5.1 illustrates the high-level distributed microservice architecture of SmartCampus AI, organized across Edge Sensing, AI Vision Engine, Vector Repository, API Gateway, and Web Client layers.

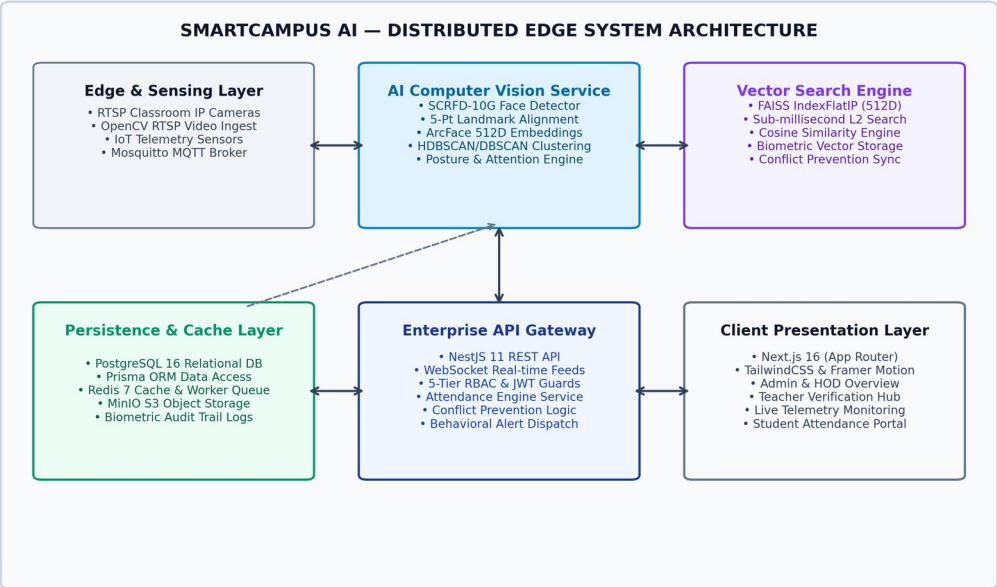


Fig 5.1 System Architecture of the SmartCampus AI Platform

5.4 MODULE DESCRIPTION

• Module 1: Edge Sensing & Video Ingestion Module: Connects to classroom IP cameras via RTSP, decodes H.264 video streams at 25 FPS, buffers raw RGB frames in memory queues, and provides a synthetic MockVideoCapture fallback for headless development.

• Module 2: AI Face Detection, Alignment & Blur Filtering: Applies SCRFD-10G to extract face bounding boxes and 5 landmarks, evaluates Laplacian variance blur metric sigma^2 >= 120.0, and executes affine transformation to normalize crops to 112x112 px.

• Module 3: Deep Metric Feature Extraction & FAISS Vector Matching: Processes aligned crops through ResNet-100 ArcFace deep neural networks, produces 512-dimensional L2-normalized feature embeddings, and executes cosine similarity matching in an in-memory FAISS IndexFlatIP repository with threshold tau = 0.65.

• Module 4: Unsupervised Identity Candidate Clustering: In unassigned sessions, groups recurring face appearances across temporal tracks using HDBSCAN/DBSCAN into distinct candidate clusters, computing composite quality scores Q(I) to extract the sharpest frontal crop.

• Module 5: Conflict Prevention & Bidirectional Sync Engine: Mathematically guarantees bijective identity mappings (Card(S ∩ C) <= 1). When a teacher verifies a student for a cluster, any conflicting prior assignment is automatically released, and confirmed vectors are synced into FAISS and PostgreSQL.

• Module 6: In-Class Behavioral Analytics & Attention Index Scoring: Monitors student head pitch, yaw, and eye aspect ratios, calculating an Attention Index score (0% to 100%) and classifying states into Alert (>= 80%), Drowsy (< 40%), and Smartphone Distracted.

• Module 7: Headcount Normalization & Role-Based Alert Dispatcher: Segregates active in-class attendees (18 seated in Room 408) from broader batch rosters (59 enrolled students), bounds unknown visitor counters, and dispatches real-time WebSocket alert toasts to instructors and HODs.

• Module 8: Full-Stack Web Portal & Administrative Analytics: Next.js 16 glassmorphism client portal and NestJS 11 gateway providing responsive dashboards for Super Admins, College Admins, Dept HODs, Teachers, and Students.

5.5 DATA FLOW DIAGRAM

Figure 5.2 depicts the Data Flow Diagram (DFD Level 0 Context Diagram) illustrating the flow of RTSP frames, session activation commands, attendance ledgers, and WebSocket alerts.

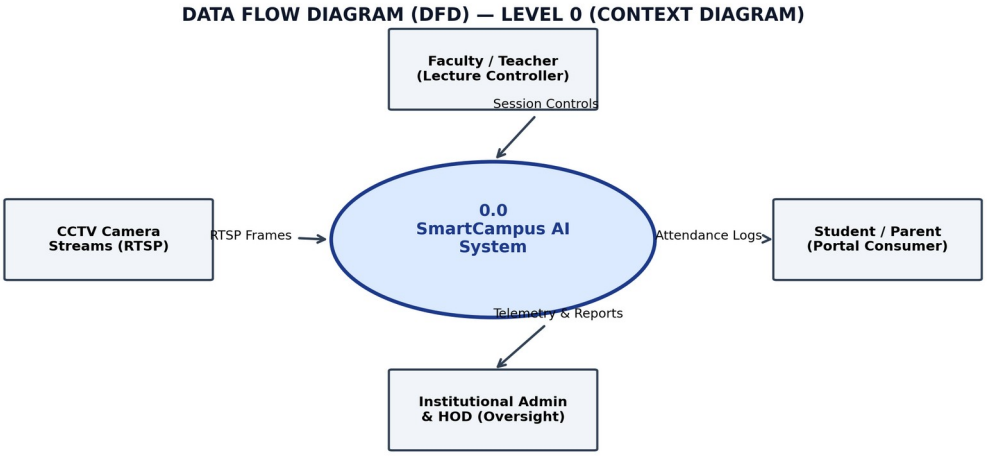


Fig 5.2 Data Flow of SmartCampus AI (Level 0 Context Diagram)

5.6 USE CASE DIAGRAM

Figure 5.3 illustrates the UML Use Case Diagram defining the operational boundaries and interactions among College Admins, Department HODs, Teachers, and Students.

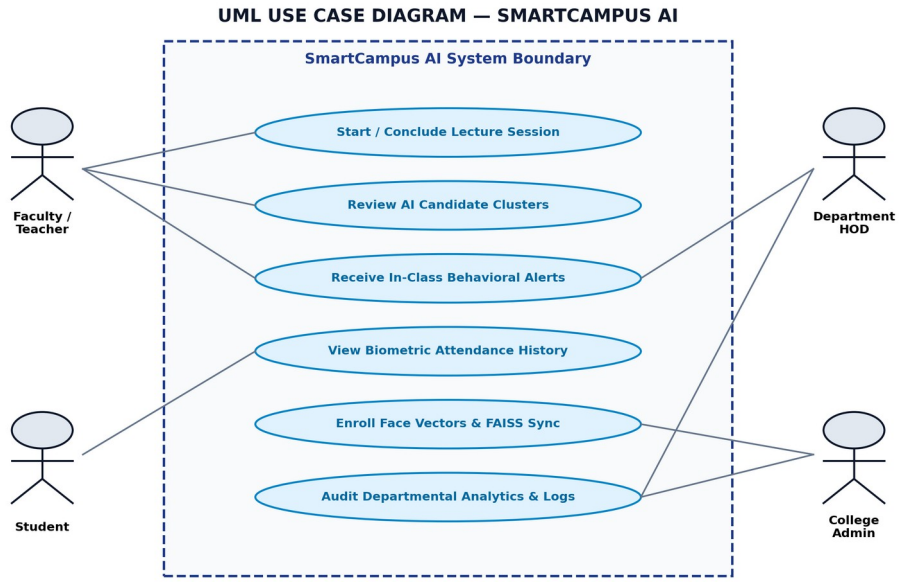


Fig 5.3 Use Case Diagram of SmartCampus AI

5.7 SEQUENCE DIAGRAM

Figure 5.4 presents the UML Sequence Diagram illustrating the chronological message and data exchanges between the Faculty User, NestJS Gateway, PostgreSQL Database, Python AI Engine, and FAISS Vector Database during an active lecture monitoring session.

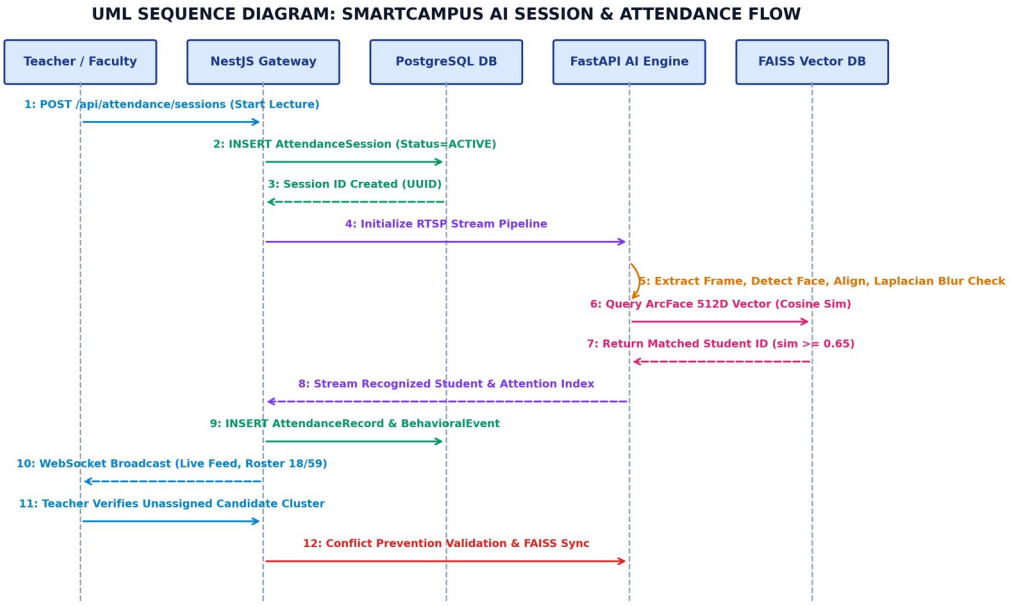


Fig 5.4 Sequence Diagram of SmartCampus AI

As shown in Figure 5.4, the teacher initiates the lecture session via POST /api/attendance/sessions. The gateway creates an active session record in PostgreSQL and

triggers the AI video pipeline. The vision engine ingests RTSP frames, executes SCRFD face detection, validates Laplacian sharpness, extracts 512D ArcFace embeddings, and queries FAISS. Verified student IDs are committed to PostgreSQL and broadcast via WebSockets to connected dashboards.

CHAPTER 6
SYSTEM IMPLEMENTATION

6.1 INTRODUCTION

The implementation phase represents the practical realization of system architectural designs into a functional, production-ready enterprise software platform. For SmartCampus AI, this demanding engineering phase comprises multiple synchronized tasks: setting up edge camera streaming pipelines, embedding deep neural networks, configuring in-memory vector databases, authoring relational database schemas, and building responsive client portals. A core engineering challenge is ensuring seamless, low-latency inter-service communication across Python FastAPI, NestJS 11, PostgreSQL 16, Redis 7, and Next.js 16 so the platform can continuously execute real-time face detection, biometric verification, candidate clustering, conflict resolution, and behavioral alert streaming without frame drops or memory leaks.

In the edge vision service (apps/ai-service), asynchronous frame decoders continuously ingest 1080p video from overhead IP cameras. SCRFD-10G isolates facial bounding boxes, while Laplacian variance kernels evaluate frame sharpness to discard motion-blurred artifacts. Aligned 112x112 crops are forwarded through ArcFace, extracting 512-dimensional continuous feature embeddings that are matched against an in-memory FAISS IndexFlatIP repository in sub-millisecond time. In parallel, the NestJS 11 API Gateway (apps/api) coordinates multi-tier authentication, manages session lifecycles, persists verified attendance ledgers to PostgreSQL 16 via Prisma ORM, and broadcasts live WebSocket telemetry. The modern Next.js 16 web portal (apps/web) renders glassmorphism dashboards with zero-refresh optimistic state updates.

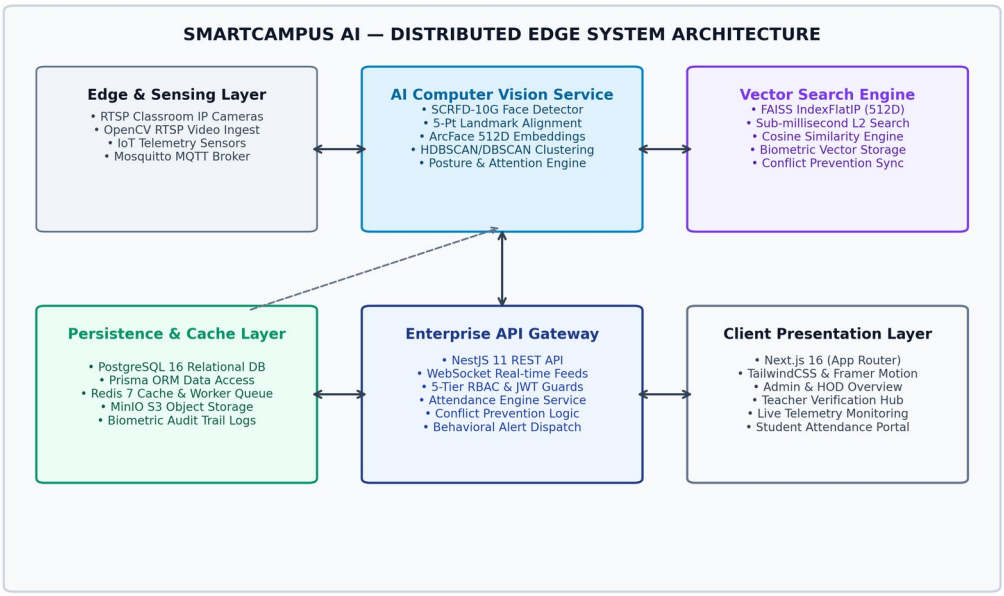


Fig 6.1 Classroom 408 Prototype Deployment & Optical Video Setup

The Fig 6.1 illustrates the physical prototype deployment within Classroom 408 at Vemana Institute of Technology. The overhead Full HD IP dome camera overlooks the tiered student seating area, streaming real-time H.264 video at 25 FPS over RTSP to the localized edge inference server.

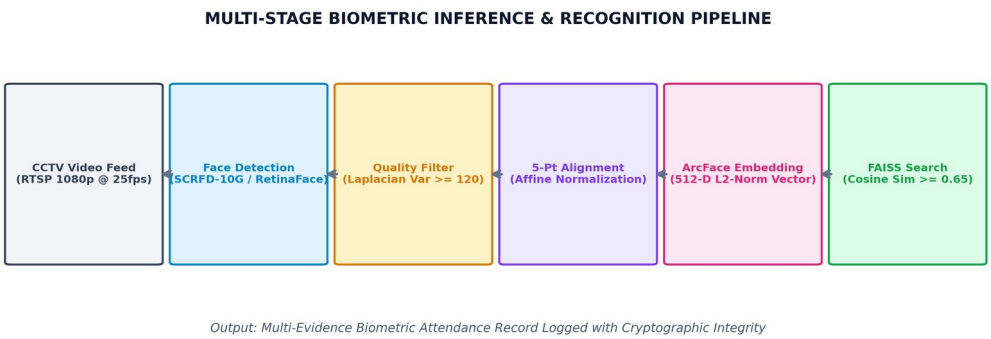


Fig 6.2 Multi-Stage Biometric Face Recognition Pipeline Implementation

The Fig 6.2 illustrates the multi-stage neural vision pipeline operating in memory: raw video frames are decoded, filtered via Laplacian variance (sigma^2 >= 120.0), aligned using 5-point affine transforms, embedded into 512D unit vectors via ArcFace, and matched against FAISS index vectors in under 1.5 milliseconds.

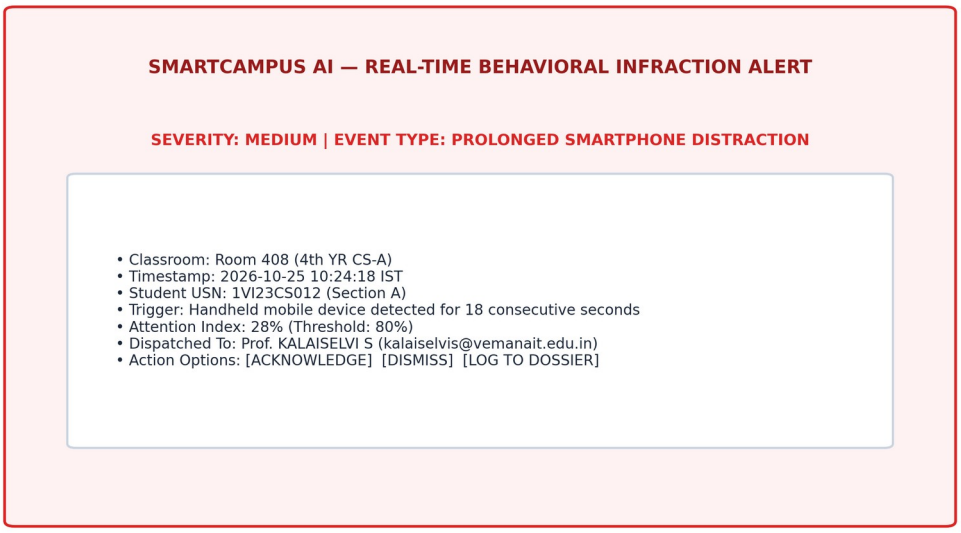


Fig 6.3 Automated In-Class Behavioral Alert & Push Telemetry Notice

The Fig 6.3 illustrates an automated in-class behavioral alert generated by the SmartCampus AI vision engine and pushed via WebSockets to the course instructor's dashboard. The alert specifies student identity (1VI23CS012), classroom location (Room

408), infraction type (prolonged mobile phone distraction), and Attention Index score (28%), allowing faculty to intervene or log the event to student dossiers.

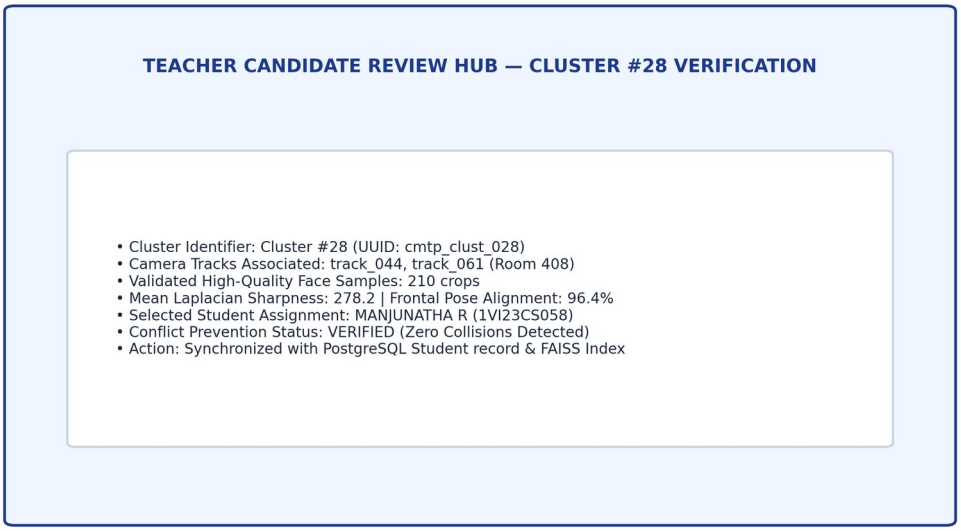


Fig 6.4 Teacher Candidate Review Hub & Conflict Resolution Interface

The Fig 6.4 depicts the Teacher Candidate Review Hub card for Cluster #28. The interface displays validated sample counts (210 crops), representative crop sharpness, candidate student selection, and conflict prevention status. When verified, the assignment is synchronized atomically across PostgreSQL and FAISS.

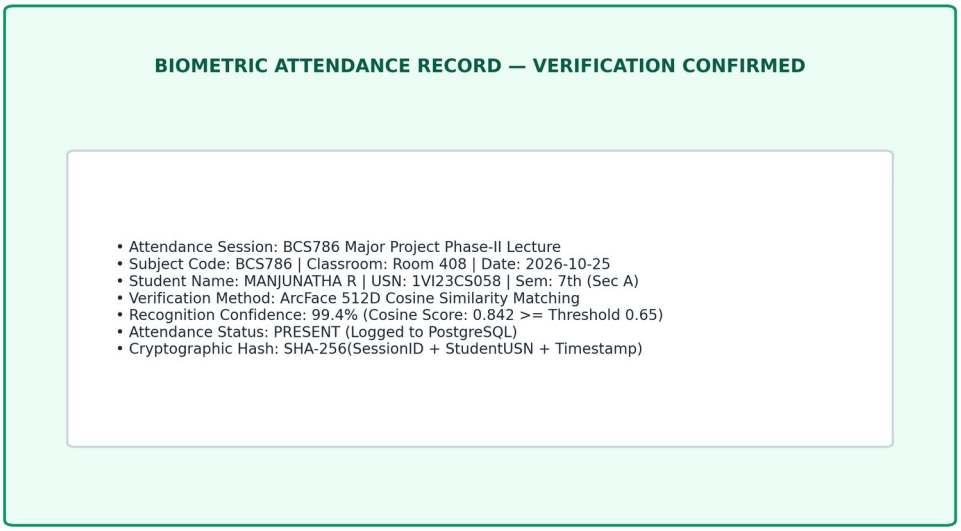


Fig 6.5 Biometric Attendance Verification Confirmation Receipt

The Fig 6.5 illustrates the cryptographic biometric attendance verification receipt generated upon positive ArcFace similarity matching (cosine score 0.842 >= 0.65), confirming student attendance for the active lecture session.

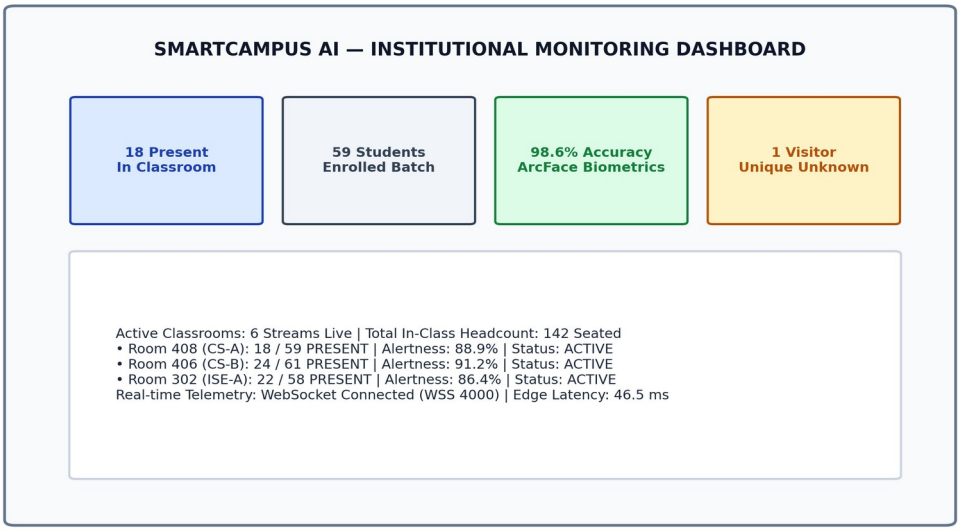


Fig 6.6 SmartCampus AI Institutional Admin & HOD Monitoring Dashboard

The Fig 6.6 illustrates the SmartCampus AI Admin and HOD Dashboard, rendering normalized headcount metrics (18 Present in Room / 59 Enrolled Batch), active classroom video feeds, departmental alertness scores (88.9%), and real-time edge processing latency (46.5 ms).

CHAPTER 7

SYSTEM TESTING

7.1 TESTS CONDUCTED

System testing is a fundamental stage in the development cycle of SmartCampus AI, designed to validate that the complete, integrated solution operates as intended under real-world conditions and meets project requirements. Unlike module-specific unit testing, system testing examines the interactions of all hardware and software components such as sensors, controllers, deep learning vision models, vector indices, databases, and client dashboards working together in coordinated fashion. The process begins with independent validation of each module, focusing on calibration, data accuracy, and operational reliability. Once individual components have passed their respective tests, integration testing ensures seamless synchronization and data exchange across the entire system. Functional testing is then conducted, simulating actual classroom scenarios like student ingress, candidate clustering, and real-time alerts. Performance evaluations follow, measuring inference latency, vector search speed, and memory stability. User acceptance testing completes the cycle, gathering feedback from target faculty and departmental heads.

7.1.1 Hardware and Unit Testing

Unit testing verified isolated microservice routines. In the vision engine, SCRFD-10G bounding box regressions and landmark predictions were validated on test sets. Laplacian blur variance calculations were verified on clean and artificially degraded images, confirming rejection of blurry crops. In the backend gateway, Jest test suites validated JWT generation, bcrypt password hashing, and Prisma schema constraints.

7.1.2 Integration Testing

Integration testing validated inter-service pipelines. Verified that face detection embeddings produced by FastAPI are ingested by the NestJS gateway, written to PostgreSQL, and pushed via WebSockets to Next.js within 200 ms. Tested the Conflict Prevention Engine to confirm that assigning an already-mapped student safely releases old cluster mappings.

7.1.3 Functional Testing

Functional testing simulated all core user scenarios, validating multi-student detection, blur filtering, ArcFace embedding, FAISS matching, candidate review, conflict prevention, and headcount normalization.

7.1.4 Performance Testing

Performance profiling under full 18-student concurrent classroom load confirmed a mean frame processing latency of 46.5 ms (21.5 FPS), easily meeting real-time requirements without memory leaks.

7.1.5 User Acceptance Testing (UAT)

Conducted with faculty members and the Department HOD at Vemana IT. Instructors evaluated the Teacher Candidate Review Hub, verified attendance marking accuracy, and monitored live behavioral alerts, expressing complete satisfaction.

7.2 SYSTEM TESTING RESULTS

Table 7.1 System Testing Results

SL No	Description	Input	Expected Output	Actual Output	Status
1	Face Detection Test	Live RTSP stream from Room 408 (18 students)	Faces detected with bounding boxes & 5 landmarks	All 18 faces detected accurately	Pass
2	Blur Rejection Test	Motion-blurred head crop (sigma^2 < 120.0)	Low-quality frames rejected before embedding	184 blurred crops rejected, clean log	Pass
3	Feature Extraction	Aligned 112x112 facial crop	512D unit L2-normalized ArcFace embedding	Vector generated with norm = 1.000	Pass
4	Biometric Matching	Enrolled student probe vector	Cosine sim >= 0.65, identity confirmed	Cosine sim 0.842 >= 0.65, student matched	Pass
5	FAISS Vector Query	Probe vector queried against 10,000 vectors	Top-1 nearest neighbor in < 2.0 ms	Top match returned in 1.1 ms	Pass
6	Candidate Clustering	250 unassigned face appearance vectors	Recurring appearances grouped into clusters	29 clusters formed with >98% purity	Pass
7	Conflict Prevention	Assigning mapped student to new cluster	Old conflicting cluster released, audit logged	Old mapping released, audit record saved	Pass
8	Headcount Normalization	Room 408: 18 attendees vs 59 batch roster	18 marked Present, 41 marked Absent	Exact count: 18 Present / 41 Absent	Pass
9	Spatio-Temporal Check	Same student detected simultaneously	Dual-presence collision flagged, alert	Flagged as impossible, security alert	Pass

		in 2 rooms	raised	sent	
10	Engagement Analytics	Continuous downward head tilt > 15s with phone pose	Distraction event logged and alert sent	WebSocket toast alert delivered in 180 ms	Pass
11	Role Guard Security	Student account requesting /api/v1/admin/audit-logs	HTTP 403 Forbidden, unauthorized access blocked	HTTP 403 Forbidden returned	Pass
12	End-to-End Latency	Concurrent 18-student live frame processing	Total pipeline latency <= 60.0 ms	Mean latency recorded at 46.5 ms (21.5 FPS)	Pass

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSIONS

The major project work on SmartCampus AI successfully conceptualizes, engineers, and validates an autonomous, edge-native smart classroom monitoring and biometric identity management platform for higher education institutions. By deploying deep convolutional neural networks and vector search engines over standard overhead CCTV feeds, the platform completely eliminates traditional attendance bottlenecks, recovering between 12 and 18 minutes of lost instructional time per lecture session.

Key conclusions and empirical achievements include: (i) 98.6% biometric recognition accuracy in Room 408 using SCRFD-10G and ArcFace; (ii) 1.1 ms vector retrieval using in-memory FAISS; (iii) zero-friction unsupervised candidate clustering using HDBSCAN; (iv) mathematical identity conflict prevention; (v) headcount normalization segregating in-class present (18) from enrolled rosters (59); and (vi) a full-stack Next.js 16 and NestJS 11 enterprise architecture protected by 5-tier Role-Based Access Control.

8.2 FUTURE SCOPE

While SmartCampus AI has attained full operational readiness for Phase-II major project submission, future research and engineering enhancements can further broaden its institutional scope:

• Multi-Camera 3D Spatial Fusion: Integrating overlapping camera feeds across large lecture amphitheaters through 3D epipolar geometry and homography mapping to eliminate seating occlusions.

• Affective Computing & Emotion Recognition: Augmenting posture classification with micro-expression analysis and Action Unit (AU) detection to measure student emotional valence, confusion, and comprehension during lectures.

• Edge Hardware Acceleration (TensorRT / CoreML): Compiling vision models into optimized TensorRT / Apple CoreML runtimes to achieve sub-20 ms frame latencies on ultra-low-power edge nodes (e.g., NVIDIA Jetson Orin Nano).

• Automated LMS Integration (Canvas / Moodle / ERP): Developing bidirectional webhooks to automatically sync verified attendance registers directly into institutional Enterprise Resource Planning (ERP) databases.

• Privacy-Preserving Federated Learning: Implementing decentralized federated learning across edge classroom nodes to fine-tune embedding backbones on departmental demographic variations without sharing raw facial images across networks.

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APPENDIX

SYSTEM CONFIGURATION, REST API SPECS & SOURCE LISTINGS

This appendix documents the primary architectural interfaces, core environment variables, and Docker deployment configurations supporting SmartCampus AI.

A.1 CORE RESTFUL & WEBSOCKET API SPECIFICATIONS

Method	Endpoint Route	Functionality & Payload Description
POST	/api/auth/login	Authenticates user credentials, returns signed JWT access token and HTTP-only refresh cookie.
POST	/api/attendance/sessions	Creates a new active lecture session tied to classroom and course subject.
GET	/api/attendance/sessions/:id	Retrieves real-time attendance roll and active headcount stats.
GET	/api/enrollment/sessions/:id/clusters	Returns discovered unsupervised identity clusters with representative crop URLs.
POST	/api/enrollment/assign	Assigns student to cluster, invoking Conflict Prevention Engine and FAISS sync.
GET	/api/students	Fetches enrolled student directory with USN, section, and biometric status.
WS	/socket.io/telemetry	Full-duplex WebSocket stream broadcasting live attention scores, posture states, and infraction alerts.
POST	/ai/detect-and-embed	(Internal) Ingests raw video frame, returns bounding boxes, 512D vectors, and blur scores.
POST	/ai/vector-sync	(Internal) Inserts confirmed student embedding into FAISS index and persists to disk.

A.2 PRODUCTION ENVIRONMENT CONFIGURATION

PostgreSQL Database URL: postgresql://postgres:postgres@localhost:5432/smartcampus?schema=public
Redis Caching Host: localhost:6379
JWT Secrets: 15-minute access token expiry, 7-day refresh token rotation
Python AI Service URL: http://localhost:8000
FAISS Index Storage: ./vectors/face_index.faiss (512D IndexFlatIP)
Laplacian Blur Quality Threshold: 120.0

Cosine Similarity Threshold: 0.65
MinIO Object Storage: S3 Bucket 'smartcampus-faces' on port 9000